

Graphs & Transformations

ANSWER KEY



Sketching quadratic graphs

- Find the vertex and y -intercept of $y = x^2 - 6x + 5$.
Vertex x -coordinate: $\frac{6}{2} = 3$. $y = 9 - 18 + 5 = -4$. Vertex: $(3, -4)$. y -intercept (at $x = 0$): $y = 5$.
- Find the vertex and x -intercepts of $y = x^2 - 2x - 8$.
Vertex x -coordinate: $\frac{2}{2} = 1$. $y = 1 - 2 - 8 = -9$. Vertex: $(1, -9)$. Factoring: $(x - 4)(x + 2) = 0$, so the x -intercepts are at $x = 4$ and $x = -2$.

Translations of graphs

- The graph of $y = f(x)$ is translated 3 units right and 2 units down. Write the equation of the resulting graph in terms of f .
A translation right by 3 replaces x with $x - 3$; down by 2 subtracts 2: $y = f(x - 3) - 2$.
- The graph of $y = x^2$ is translated to have vertex $(-2, 5)$. Write its equation.
 $y = (x + 2)^2 + 5$.
- The graph of $y = x^3$ is translated 4 units left and 1 unit up. Write the equation of the resulting graph.
A translation left by 4 replaces x with $x + 4$; up by 1 adds 1: $y = (x + 4)^3 + 1$.

Stretches and reflections

- Describe the transformation that maps $y = x^2$ to $y = 3x^2$.
A vertical stretch with scale factor 3 (each y -value is tripled).
- Describe the transformation that maps $y = f(x)$ to $y = f(2x)$.
A horizontal stretch with scale factor $\frac{1}{2}$ (the graph is compressed toward the y -axis).
- Describe the transformation that maps $y = f(x)$ to $y = -f(x)$.
A reflection in the x -axis.
- Describe the transformation that maps $y = f(x)$ to $y = f(-x)$.
A reflection in the y -axis.

Combining transformations

- 10** Given $f(x) = x^2$, describe the sequence of transformations that produces $g(x) = 2(x - 1)^2 + 3$, in the order they are applied.
Starting from $f(x) = x^2$: first shift right by 1 to get $(x - 1)^2$; then stretch vertically by a factor of 2 to get $2(x - 1)^2$; then shift up by 3 to get $2(x - 1)^2 + 3 = g(x)$.
- 11** Given $f(x) = x^2$, describe the sequence of transformations that produces $h(x) = -f(x + 2) + 1$, in the order they are applied.
Starting from $f(x) = x^2$: first shift left by 2 to get $(x + 2)^2$; then reflect in the x-axis to get $-(x + 2)^2$; then shift up by 1 to get $-(x + 2)^2 + 1 = h(x)$.
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Graphs of the modulus function

- 12** Sketch $y = |x - 3|$ and state its minimum value and where it occurs.
The graph is a V-shape with vertex at $(3, 0)$. The minimum value is 0, occurring at $x = 3$.
- 13** Solve $|2x - 5| = 7$.
Either $2x - 5 = 7$, giving $x = 6$, or $2x - 5 = -7$, giving $x = -1$. So $x = 6$ or $x = -1$.
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Finding the equation of a transformed curve

- 14** A curve $y = f(x)$ has a minimum point at $(4, -2)$. State the coordinates of the minimum point on the graph of $y = f(x - 3) + 5$.
The transformation shifts the graph right by 3 and up by 5, so the minimum point moves to $(4 + 3, -2 + 5) = (7, 3)$.
- 15** A curve $y = f(x)$ has a maximum point at $(-1, 6)$. State the coordinates of the maximum point on the graph of $y = 2f(x) - 3$.
The transformation stretches vertically by factor 2 and then shifts down 3, so the x-coordinate is unchanged and the y-coordinate becomes $2(6) - 3 = 9$. Maximum point: $(-1, 9)$.
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Intersections of transformed graphs

- 16** Find the coordinates of the points where $y = (x - 2)^2 - 1$ crosses the x-axis.
Set $y = 0$: $(x - 2)^2 = 1$, so $x - 2 = \pm 1$, giving $x = 3$ or $x = 1$. Points: $(1, 0)$ and $(3, 0)$.
- 17** Find the coordinates of the points where $y = (x + 1)^2 - 4$ crosses the x-axis.
Set $y = 0$: $(x + 1)^2 = 4$, so $x + 1 = \pm 2$, giving $x = 1$ or $x = -3$. Points: $(-3, 0)$ and $(1, 0)$.
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Graphs of reciprocal functions

- 18 State the equations of the asymptotes of $y = \frac{1}{x-2} + 3$.

Vertical asymptote (where the denominator is zero): $x = 2$. Horizontal asymptote (as $x \rightarrow \pm \infty$, the fraction term vanishes): $y = 3$.

- 19 State the equations of the asymptotes of $y = \frac{3}{x+4} - 2$.

Vertical asymptote (where the denominator is zero): $x = -4$. Horizontal asymptote (as $x \rightarrow \pm \infty$, the fraction term vanishes): $y = -2$.

A well-designed word problem: modeling a bridge cable

- 20 This question has two parts. A suspension bridge cable hangs in the shape of a parabola. Measuring from the lowest point of the cable, the cable rises to a height of 20 meters at a horizontal distance of 50 meters on each side. (a) Using the lowest point of the cable as the origin, write the equation of the parabola in the form $y = ax^2$, finding a . (b) Find the height of the cable at a horizontal distance of 30 meters from the lowest point.

(a) At $x = 50$, $y = 20$: $20 = a(50)^2 = 2500a$, so $a = \frac{20}{2500} = \frac{1}{125} = 0.008$. Equation: $y = 0.008x^2$. (b) At $x = 30$: $y = 0.008(900) = 7.2$ meters.